WRONG BIRD

Game Design Document

Engine TBD | A game by Clay Buiocchi

“Penguin’s can’t fly” -Some idiot



Target Deliverable
An absurd take on rougelites that is incredibly replayable and easy to pick up. I want the player to be smiling the whole time!

Overview

Tagline

Raise, train, and launch penguins in order to prove that these birds aren't flightless after all!

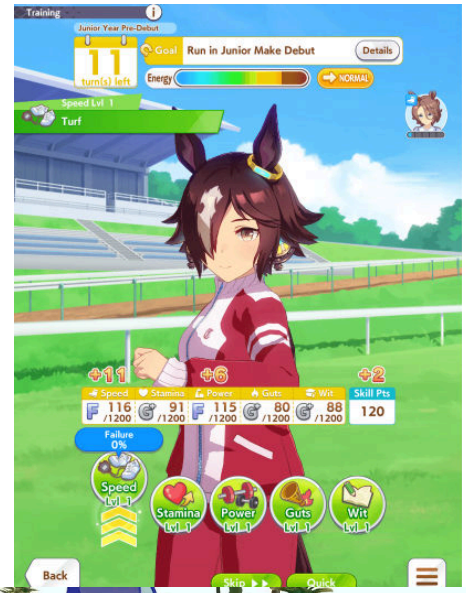
Genre and Theme

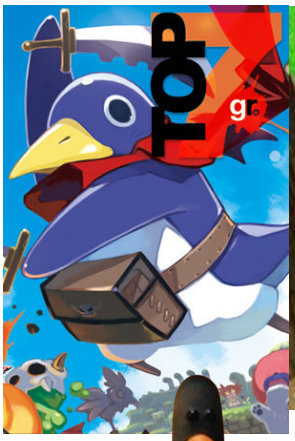
- **Genre:** Distance Launcher / Raising Simulator Rougelite
- **Subgenre:** Rougelite progression with training, shop, evolution, and breeding systems
- **Theme:** Absolute absurdity — penguins conquering physics through sheer willpower
- **Aesthetic:** Bright GameCube inspired 3D aesthetic

Target Audience and Platform

- **Audience:** Casual to mid-core players who enjoy watching number go up, iterative gameplay
- **Comparable Fans:** Fans of games like Umamusume and the Chao Garden from the Sonic Adventure games, as well as Mewgenics
- **Platform:** PC/Mac, eventual port to console and mobile
- **Input:** Keyboard/mouse or controller

Moodboard


Training
Junior Year Pre-Debut
11
turn(s) left
Goal
Run in Junior Make Debut
Details
Energy
NORMAL
Speed Lvl 1
Turf
+11
+6
+2
Speed
Stamina
Power
Guts
Wit
Skill Pts
116 /1200
91 /1200
115 /1200
80 /1200
88 /1200
120
Failure 0%
Speed Lvl 1
Stamina Lvl 1
Power Lvl 1
Guts Lvl 1
Wit Lvl 1
Back
Skip
Quick

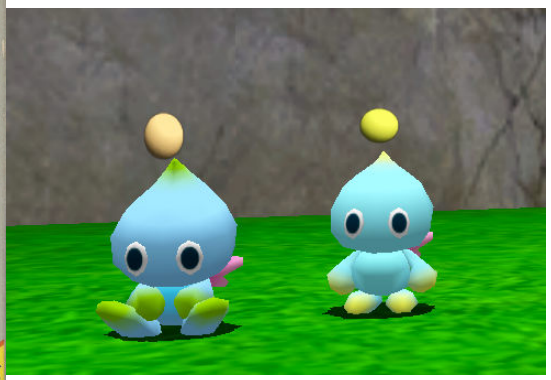

TOP


7 / 8
TIME 00:10:88
C.R. 00:17:78
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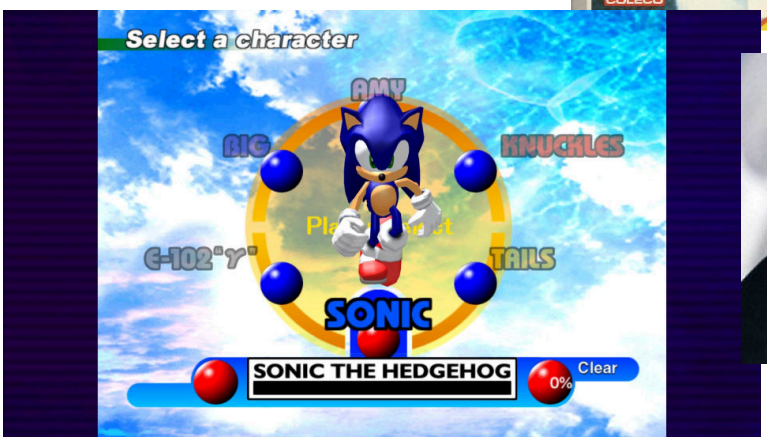

SCORE
20484
051
TIME
BANANA(S)
008 / 100
22 mph
FLOOR 3




ANTARCTIC ADVENTURE
FREE MAP OF ANTARCTICA INCLUDED!
FOR USE ONLY WITH ADAM FAMILY COMPUTER SYSTEM
COLECO






Select a character
AMY
BIG
KNUCKLES
TAILS
SONIC
SONIC THE HEDGEHOG
Clear
0%



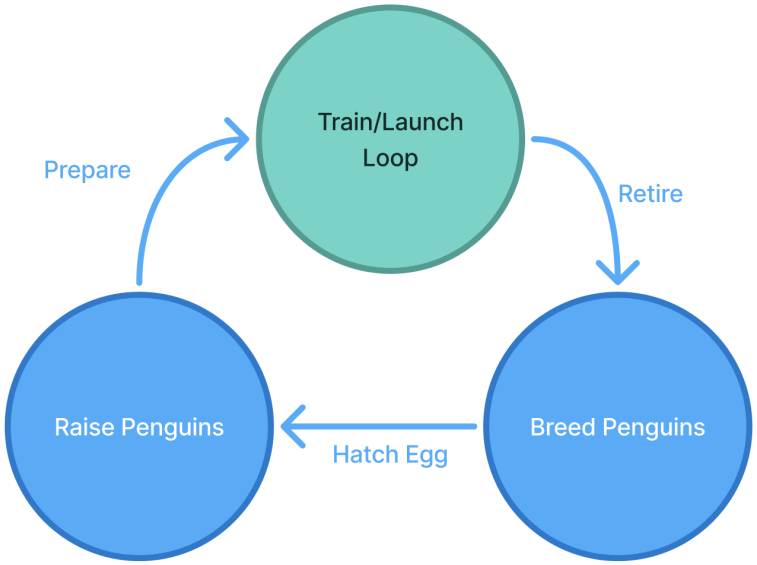


Competitive Landscape

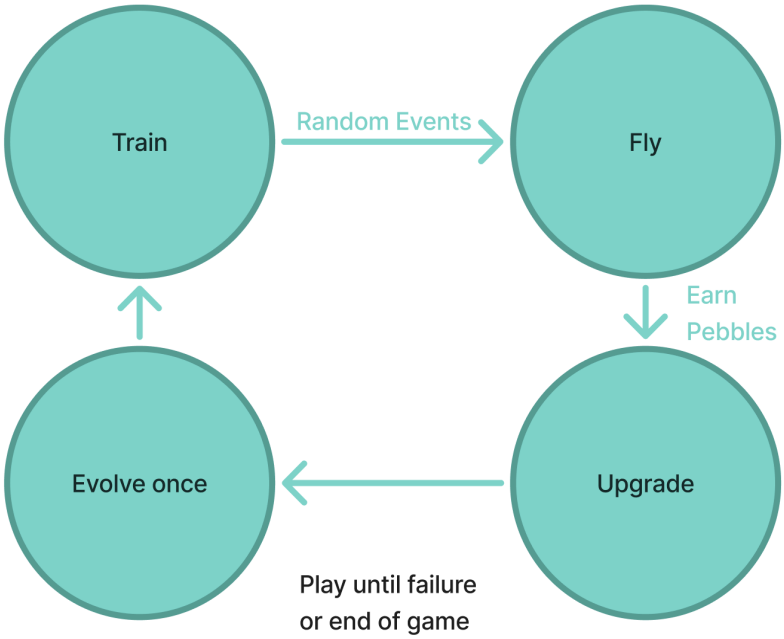
Game	Platform	Similar To	Key Difference
Umamusume: Pretty Derby	Mobile	Training loop, breeding	Gatcha mechanics
Burrito Bison: Launcha Libre	Mobile	Launch + shop	Relies on ads
Chao Garden	Dreamcast/Gamecube	Stat-training, automated races	Side game
Wrong Bird	PC/Mac/Console/Mobile	All of the above	One time purchase, full game

Gameplay

Core Loop



Training/Launch Loop



Narrative

WIP

Mechanics

Training Phase (X Days)

Each day, the player chooses one training action from a menu. Each training costs energy and has a chance to fail based on exhaustion level.

Training	Stat Gain	Energy Cost	Fail Chance
Waddle	+5 WDL	-	1% + fatigue
Flap/stamina	+5 FLP	-X	1% + fatigue
Glide	+5 GLD	-X	1% + fatigue
Slide	+5 SLD	-X	1% + fatigue
Special	+5 SPC	-X	1% + fatigue
Rest	none	+X	0%

Random Events

Each day has a X% chance to trigger a random event that modifies training conditions:

- Sunny Day — energy costs reduced by 5
- Blizzard — fail chance increased by 15%
- Albatross Race — random race/skill-check, wins give a bonus, failures give nothing
- Penguin Visit - one of your previous penguins will train your current penguin in a specific stat, small chance to unlock a special ability
- Special penguin visit — +5 to all stats
- Good Sleep — +20 energy bonus
- Lucky Day — double gains, zero fail chance
- Exhausted — forces rest (triggers below 20 energy)

Launch Phase

After Day X, the player enters the launch phase. The penguin waddles and jumps off a cliff and travels across a procedurally scrolling environment.

- Pre-launch: Mash A to fill a power bar. Adds up to +X% to base waddle speed
- In-flight: Flap drains over time; when empty, glide stat determines speed decay
- Landing: Penguin slides on the ice; slide stat and special abilities determine stopping distance
- Altitude: Staying airborne earns altitude bonus
- Special: Reaching X altitude triggers a “Space Penguin!” event

Abilities & Shop

After a successful launch, the player can visit the ability shop and spend pebbles on abilities that activate automatically at set distances during flight.

Name	Category	Pebble Cost	Trigger	Description
Waddle Sprint	Waddle	200	Launch	Greatly increases launch speed
Second Wind	Flap	800	Run out of stamina	When penguin runs out of stamina, give ⅓ of stamina back
Thermal Ride	Glide	500	Reach x miles	Lift penguin at 50 miles
Super Slide	Slide	600	Landing (after any bounces)	Accelerate on ice for 3 seconds
Bounce	Special	1,100	Landing	Bounce on landing

Biomes

The penguin will travel through the following biomes, each with their own challenges:

Biome	Distance	Note
Antarctica	0 mi	Home
Atacama Desert	~1,400 mi	High headwinds
Amazon Rainforest	~2,600 mi	Humidity drains flap faster
Caribbean/Gulf of Mexico	~3,800 mi	Long stretch with islands, unpredictable weather
American South	~5,200 mi	Swamps = slower slide
Appalachians	~6,200 mi	Turbulence

Biome	Distance	Note
New York City	~7,000 mi	

Evolution

At a certain point during the training launch loop, your penguin will evolve. Evolutions are lightly based off of real life penguin subspecies.

Name	Dominant Stat	Passive	Visual	Note
Emperor	Waddle	Maintains top speed for longer	Emperor penguin	
King	Flap	Flaps are more powerful, conserves stamina	King penguin	
Gentoo	Slide	Less slide friction on non-ice	Gentoo penguin	
Chinstrap	Glide	Slower descent	Chinstrap penguin	
Rockhopper	Special	Increases bounce distance	Rockhopper penguin	

Breeding System

When a penguin completes their run, they retire and join the permanent colony. Once you have two retired penguins, you can breed them to produce an egg.

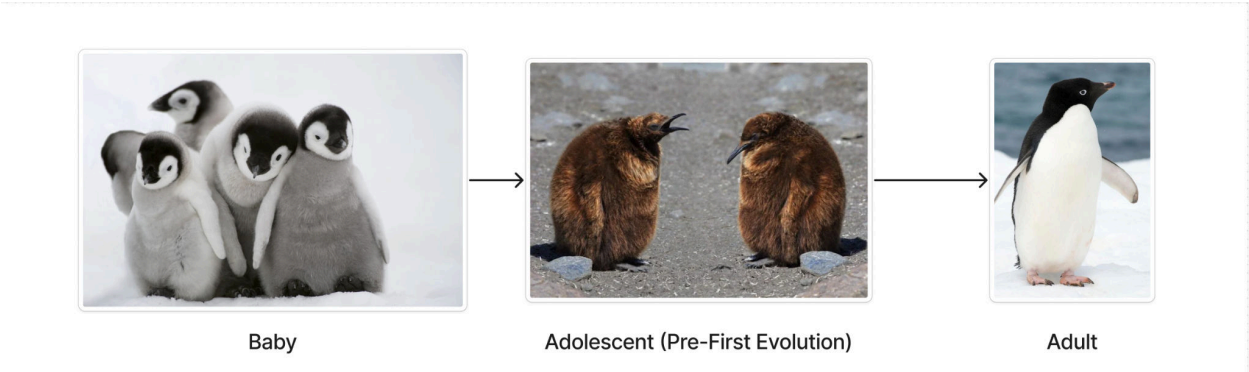
Breeding is the primary long-term progression mechanic. Each generation trends stronger than the last, but never guarantees perfection — the wild rolls and trait system ensure every penguin feels like their own individual.

Stat Inheritance

Each stat is inherited independently. For each stat, the chick rolls a value between the two parents' values, weighted to slightly favor the higher parent. This means good breeding always trends upward without guaranteeing the maximum. Additionally, one random stat receives a wildcard roll modifier of ±5, applied after the weighted roll.

First Evolution

After a certain amount of time, your penguin will go through puberty. It will become very fluffy and have brown feathers, similar to their awkward real life adolescence. Eventually, a cutscene will play, and the brown feathers will fall off to reveal your adult penguin.



Penguin Rarity System

Every penguin has a rarity tier that determines their color palette, visual effects, and a small exclusive trait. Common is the default for all hatchlings — rarity upgrades only occur when specific breeding conditions are met. Rarity does not grant raw stat advantages.

Rarity Tiers

The tiers escalate from grounded-but-stylized at Common to genuinely otherworldly at Legendary. Since the whole game is about a penguin defying physics, the visual escalation feels narratively earned — the rarer the penguin, the more clearly they were meant to fly.

Tier	Color Identity	Glow / FX	Exclusive Trait
Common	Deep navy and white, vivid yellow beak	None	
Uncommon	Teal and seafoam, mint shimmer	Subtle shimmer on feathers	
Rare	Violet and electric blue, neon accents	Soft pulse glow	
Epic	Molten gold and magenta, iridescent sheen that shifts with movement	Active shimmer; particles trail during launch	

Legendary	Pure white with prismatic rainbow iridescence	Constant soft aurora glow; unique prismatic launch particle burst	

Breeding Triggers

Rarity upgrades happen partially randomly — they also require specific trait combinations in the parents. Players must understand the trait system, build toward particular combinations across generations, and get a lucky on inheritance. Not all recipes are documented in-game upfront; discovery is part of the reward.

The Breeding Log

The Breeding Log is a small in-game journal that records every breeding attempt and its outcome. When a rarity upgrade occurs, the log notes which trait combination triggered it, letting players reverse-engineer the system over time. This rewards curiosity without requiring a wiki, and gives players a sense of scientific discovery that mirrors Jay's own research.

Rarity Design Notes

- Rarity is permanent — a penguin's tier never changes after hatching.
- Rarity does not affect colony role eligibility.
- Visual FX for Epic and Legendary tiers should be designed to be visible but not distracting during the launch phase

The Colony

Retired penguins live in your colony, available for breeding. The colony holds up to 8 penguins.

Colony Cap: Penguin Roles

When your colony is full and a new penguin retires, rather than removing anyone, you assign the new retiree a permanent role. Penguins in roles leave the breeding pool but remain in the world — they are never deleted or lost.

Role	Effect	Best For
Coach	Reduces fail chance or energy cost for one stat during training, based on the penguin's highest stat at retirement	Penguins with a clear stat specialty
Expedition Member	Each day, small chance to return with a gift, rare item, or pebble bonus	Penguins with high Special or lucky trait history

Raising System

Between and during training actions, the player has lightweight interactions with their penguin through gifts, mood management, and bond events. Raising doesn't replace training — it deepens it, unlocking long-term rewards through care rather than stat optimization.

Mood

Penguins have a mood meter alongside their energy bar. Mood doesn't directly modify stats — it affects the ceiling of training outcomes. A happy penguin has a higher chance of hitting the full stat gain; a miserable one trends toward the lower end of the range.

Event	Mood Effect
Resting	+Mood
Successful training	+Small mood
Failed training	-Mood
Receiving a gift	+Mood (varies by gift)
Winning a launch	++Mood
Missing the launch goal	-Mood
Repeating the same training 3+ days in a row	-Boredom penalty

The boredom penalty is a soft incentive to rotate training types, which also happens to be better strategy. This way, the kind choice and the optimal choice align.

Gifts

Gifts are the primary tool for mood management and the gateway to unlocking latent abilities and early stat boosts. They are purchased from a secondary gift shop using Gold Pebbles, or occasionally earned through expedition returns and events.

Gifts fall into 3 categories:

Category	Examples	Primary Effect	Secondary Effect
Food	Fish, Sardine, Fancy Fish, Anchovy Cake		
Toys	Ball, Squeaky Fish, Tiny Sled, Bowtie, TV		
Training Gear	Weighted Vest, Aerodynamic Goggles, Grip Tape		

Bond Events

At certain mood thresholds and story milestones, bond events trigger — short dialogue scenes between Jay and the current penguin.

- Bond events are 3–5 lines of dialogue
- Most are pure character moments
- Some bond events may affect mood or energy level

- Bond events are logged in a viewable journal, so players can revisit their favorite moments

Stats & Resources

Penguin Stats

All stats have soft caps per prestige level. The Special stat amplifies ability effects across the board.

Stat	Effect	Base Cap (Level 1)	Role
Waddle(WDL)	Determines launch velocity	100	Core distance driver
Flap (FLP)	Effects how long penguin can flap wings	100	Airtime duration
Glide (GLD)	Speed retention when stamina runs out	100	Mid-flight efficiency/glide
Slide (SLD)	Ice friction reduction on landing	100	Landing distance
Special (SPC)	Amplifies all ability effects by 1% each	100	Ability synergy

Resources

- **Energy:** Recovered by resting; spent on training. If it drops below 20, the penguin is forced to rest the next day.
- **Pebbles:** Training/Launch currency — earned by overshooting the goal distance. Can be used to purchase abilities
- **Gold Pebbles:** Earned at the end of training/launch cycle — used to purchase cosmetics and permanent upgrades.

Goals & Difficulty Scaling

After each successful cycle, the next goal is set to X% of the last flight distance. This creates a natural difficulty ramp that scales with the player's actual performance rather than a fixed curve.

Interface

Screen Layout

- Training Screen: Inspired by Sonic Adventure character selection screen. Each option has its own animation.

- Launch Screen: Scrolling side-view with speed lines, clouds, mountains, and a space layer. HUD shows speed (m/s), stamina bar, altitude bar, minimap with flight path, and goal marker.
- Results Screens: Post-launch summary with distance, goal, overshoot, pebble gain, and contextual hints (e.g. “train stamina!” if airtime was short).

Camera & Controls

The camera follows the penguin dynamically, similar to how the camera follows chao during chao races or umas during umamusume races. There is also a fixed camera option.

Button	Training Phase	Launch Phase
W/A/S/D, Arrow Keys, Mouse, Joystick	Navigate menus	N/A
A (Button 1)	Select	Mash pre-launch (power bar); slide boost on landing
B (Button 2)	Back	N/A

Audio

Music

- I want fun, energetic, intense music during launches (Ska, EDM, sonic adventure games)
- More laid back during training segments (chill house/jungle music)
- When the penguin is idle for a certain amount of time, it will start dancing to the music. If its in a bad mood, it will still dance, but very unenthusiastically

Sound Effects

- Cartoony, but not too over the top in order to prevent the game from becoming too overstimulating

Production

Visual Aesthetics

Art is inspired by 90s/2000s era console games, especially mascot games

- Penguin: A mix of the penguins from mario 64 and the chao from sonic adventure
- Environments: Antarctic (possible other biomes)
- UI: Michiko Sakurai-influenced (Kirby Air Riders)
- Particle effects: Speed lines, ability flashes, bounce particles, screen flash on ability trigger
- Color use: Bright and colorful

Prototype Plan

Goal: Ensure the core loop is engaging enough to play for multiple cycles.

Core Loop: Train 7 rounds, then launch. Repeat.

Risks: Gameplay could become too repetitive.

Playtesting plan:

Build order

Name	Type	Training or Launch	Priority	Notes
S_PenguinStats	Struct	Both	High	Contains penguin stats
E_TrainingType	Enum	Training	High	
E_TrainingOutcome	Enum	Training	High	
E_LaunchState	Enum	Launch	High	
S_TrainingOption	Struct	Training	High	
E_MoodState	Enum	Training	Medium	
E_RandomEventType	Enum	Training	Medium	2-3 random events to start
S_RandomEvent	Struct	Training	Medium	

BP_WrongBirdGameInstance	Game Instance	Both	High	
BP_WrongBirdGameMode	Game Mode	Both	High	
L_Training	Level	Training	High	Training scene
L_Launch	Level	Launch	High	Launch scene
BP_PenguinActor	Actor	Training	High	Player penguin
ABP_Penguin	Animation	Both	High	
BP_CameraRig_Training	Actor	Training	High	
WBP_StatBar	Widget	Training	High	
WBP_EnergyBar	Widget	Training	High	
WBP_TrainingOption	Widget	Training	High	5 stats, including failure rate
WBP_TrainingMenu	Widget	Training	High	Contains vertical box of training options
BP_RandomEventManager	Actor	Training	Medium	
WBP_RandomEventPopup	Widget	Training	High	
SEQ_Waddle_Success	Sequencer	Training	Low	
SEQ_Waddle_Failure	Sequencer	Training	Low	
SEQ_Flap_Success	Sequencer	Training	Low	
SEQ_Flap_Failure	Sequencer	Training	Low	
SEQ_Glide_Success	Sequencer	Training	Low	
SEQ_Glide_Failure	Sequencer	Training	Low	
BP_Penguin	Actor	Launch	High	
BP_CameraRig_Launch	Actor	Launch	High	
BP_GoalMarker	Actor	Launch	High	
WBP_PowerBar	Widget	Launch	Medium	
WBP_LaunchHUD	Widget	Launch	High	

WBP_Results	Widget	Launch	High	

Project Scope

This project is a portfolio piece for the Circuit Stream Game Design Bootcamp.

Area	Scope	Status	Notes
Core Loop	Training + Launch + Pebbles	Pico 8 prototype complete	7 day cycle
Abilities	20+ abilities	Pico 8 prototype complete	All trigger + effect types implemented
Evolution System	~20 evolutions	Not Started	Stat scaling
Breeding System		Not Started	
Raising/Gift System	80+ gifts	Not Started	
Weather System	6 weather types	Pico 8 prototype complete	Affects glide, slide, and wind
Events System	6 daily event types	Pico 8 prototype complete	Affects training outcomes
Visual Effects	Particles, speed lines, flashes	Pico 8 prototype complete	
Audio	Music and sound effects	Not started	
Polish	UI balance, game feel	Ongoing	

(think about randomized passives/ abilities, unique passives)